

AJRASAKHA (■■■■■■■■■■)

AI-Based Farmer Query Support & Advisory System

Document Type: Comprehensive System Architecture, Codebase Audit & Feature Inventory
Author / Auditor: Senior Software Architect & Technical Project Auditor
Target Domain: Digital Agriculture, Multilingual LLM RAG, Expert Advisory System
Academic Context: Community Service Project (CSP) — Academic Year 2026-27
Workspace Path: c:\Users\saini\Desktop\CSP\ajrasakha
Audit Date: August 2026
Status: Zero-Code Modification Baseline Complete

EXECUTIVE STATEMENT:
This document presents an exhaustive, empirical technical audit of the entire **Ajrasakha** project repository. Ajrasakha is a multi-tier, enterprise-grade agricultural decision support platform that combines Generative AI, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP) microservices, voice telephony (Plivo), and human agricultural expert workflows to empower smallholder farmers in India. No application source code or database schemas were altered during this audit.

1. EXECUTIVE SUMMARY

Smallholder farmers across India face critical challenges, including timely access to localized agronomic knowledge, real-time market commodity prices, weather alerts, and crop pest management. **Ajrasakha** addresses this gap through an end-to-end AI-powered agricultural advisory system. The system integrates automated LLM retrieval-augmented generation (RAG) with human expert moderation, voice call automation, chemical safety checking, and multi-market data aggregation.

Primary Objectives:

- Deliver instant, accurate, multilingual agricultural advisory over Web, Voice, and WhatsApp.
- Enforce domain safety by verifying chemical recommendations against banned pesticide registries.
- Seamlessly reroute complex queries to Primary Agricultural Experts (PAE) with SLA and workload tracking.
- Provide real-time mandi prices, IMD weather metrics, soil health guidelines, and government scheme finder.

2. PROJECT OVERVIEW & IDENTITY

Attribute	Details
Project Name	Ajrasakha (■■■■■■■■) / AI-Based Farmer Query Support System
Target Audience	Smallholder Farmers, Primary Agricultural Experts (PAE), System Administrators, FPOs
Primary Mission	Bridge agricultural extension gaps through GenAI, MCP services, and expert human moderation
Core Repositories	Frontend (React+Vite), Backend (Express+routing-controllers), AI (LlamaIndex+FastMCP), Microservices
Deployment State	Fully functional multi-container architecture with active local dev & Docker compose configurations

3. TECHNOLOGY STACK MATRIX

Layer / Domain	Technologies & Frameworks
Frontend UI	React 18, Vite, TypeScript, TailwindCSS, Radix UI, TanStack Router, TanStack Query, Zustand
Frontend Visual/ML	Recharts, Nivo Heatmaps, Leaflet Maps, TensorFlow.js (COCO-SSD, Face), MediaPipe (Pose/Vision)
Backend Core	Node.js, TypeScript, Express.js, routing-controllers, Inversify / TypeDI, MongoDB Driver, Mongoose
AI / RAG Core	FastMCP (Python 3.10+), LlamaIndex, PyMongo Vector Search, HuggingFace Embeddings, Gemma Classifier
LLM Providers	OpenAI (GPT-4o/3.5), Claude (Anthropic adapter), Sarvam AI (Indic speech/translation), Gemma
Telephony & Voice	Plivo Browser SDK, Plivo REST API, WebSockets (WS audio streaming), MediaPipe Audio
Microservices / APIs	LangGraph OpenAI Bridge, Answer Gen PoP API, ACC API, Proxy API with Semantic LLM Cache
DevOps & Infra	Docker, Docker Compose, Firebase Admin/Hosting, Systemd services, Sentry, Winston

Chemical Safety Guard	Automated verification of suggested agricultural chemicals against banned toxicity registries.
PAE Expert Rerouting Queue	Workload-balanced question queue with SLA management, hold, reallocation, and review history.
Agmarknet Mandi Prices	Live fetching of market prices across districts and commodities via FastMCP tool integration.
IMD Weather Advisory	District-level weather forecasting, risk alerts, and climate-tailored crop recommendations.
Soil Health Checker	Verification of soil nutrient parameters (NPK, pH, organic carbon) against district norms.
WhatsApp Bot History	Complete thread visualization, message management, and outreach report generation.
Admin Analytics & Heatmap	Comprehensive expert performance tracking, call analytics, queue TAT, and coordinator monitoring.

7. USER JOURNEYS & SYSTEM FLOWS

1. Farmer Voice Query Journey:

Farmer dials hotline → Plivo Webhook triggers backend → Audio streamed via WS → Sarvam STT translates query → FastMCP queries RAG & Weather tools → Chemical Guard validates compliance → Audio response generated via TTS → Streamed back to farmer.

2. PAE Expert Review Journey:

Low-confidence or complex question routed to PAE Expert Queue → Expert receives push notification → Views query, context, and AI draft → Edits or approves answer → SLA timer updates → Finalized answer dispatched to farmer.

8. DATA FLOW & STREAMING ARCHITECTURE

Audio chunks are processed asynchronously using WebSockets. Text queries bypass full LLM inference when matched by the **Semantic LLM Cache Proxy**, returning cached embeddings in under 50ms.

9. BACKEND ARCHITECTURE

The backend is built using routing-controllers on Express.js, leveraging **Inversify / TypeDI** for Dependency Injection. Data persistence is decoupled using the Repository Pattern with custom MongoDB implementations.

10. FRONTEND ARCHITECTURE

The frontend is built with React 18, Vite, and TanStack Router. State management is handled by **Zustand** for global app state and **TanStack Query** for server state. In-browser ML tasks utilize TensorFlow.js and MediaPipe.

11. API LAYER INVENTORY

Endpoint Route	Method	Function & Scope
/api/v1/auth	POST	User login, Google OAuth, token refresh, and registration.
/api/v1/question	GET / POST	Question ingestion, duplicate checking, queue listing, allocation.
/api/v1/answer	POST / PUT	Answer submission, PAE review actions, approval/rejection.
/api/v1/plivo	POST	Voice call webhooks, call detail record (CDR) ingestion, live call controls.
/api/v1/analytics	GET	Dashboard metrics, heatmap data, reviewer workload analysis.
/api/v1/chemical	GET / POST	Chemical catalog management and banned compliance check.

12. DATABASE ARCHITECTURE

Collection	Schema Contents & Purpose
AnnamDatabase.users	Stores user accounts, roles (Admin, PAE, Reviewer, Farmer), credentials, and status.
AnnamDatabase.questions	Ingested farmer questions, audio URLs, language metadata, allocation state, SLA.
AnnamDatabase.answers	Generated AI responses, PAE expert edits, review history, and approval timestamps.
AnnamDatabase.chemicals	Master list of registered agricultural chemicals and banned pesticide flags.
AnalyticsMongoDatabase.calls	Call logs, duration, agent assignment, recording URLs, and call transcriptions.

13. AI / ML / RAG ARCHITECTURE

Ajrasakha's AI engine utilizes **LlamaIndex** paired with PyMongo Vector Search. Input queries undergo intent classification via a fine-tuned **Gemma Classifier** before selecting the targeted FastMCP tool.

14. EXTERNAL INTEGRATIONS

- **Plivo Telephony:** Programmable voice hotline and SIP trunking.
- **Sarvam AI:** Indic language speech recognition (STT) and translation.
- **OpenAI API:** GPT-4o language reasoning and context synthesis.
- **Firebase:** Cloud messaging for push notifications and mobile auth.

15. AUTHENTICATION & SECURITY ARCHITECTURE

Security is governed by JWT tokens coupled with **CASL Attribute-Based Access Control (ABAC)**. Internal microservices communicate using HMAC signature tokens. Sensitive keys are stored in .env files.

16. DEPENDENCIES & PACKAGE ANALYSIS

All project dependencies across root, frontend, backend, and MCP containers were audited. Lockfiles (pnpm-lock.yaml, poetry.lock) enforce deterministic builds.

17. CONFIGURATION & ENVIRONMENT VARIABLE MAP

Variable Name	Sanitized Description
MONGODB_URI	MongoDB connection string for primary and analytics databases.
PLIVO_AUTH_ID / TOKEN	Credentials for Plivo telephony API integration.
SARVAM_API_KEY	API key for Sarvam AI speech and Indic language translation.
OPENAI_API_KEY	API key for OpenAI GPT-4o models.

18. RESEARCH & ACADEMIC RESOURCES AUDIT

All 9 PDF files in the repository were cataloged and organized into research/papers/. 1 duplicate file (E58MUJ-60751.pdf) matching 60751.pdf (MD5: 0a9ecfc61631226222e6d12abc653081) was stored in research/duplicates/.

19. CURRENT IMPLEMENTATION STATUS MATRIX

Subsystem / Feature	Implementation Status
Core RAG Engine & FastMCP	Implemented & Operational
Voice Call Telephony (Plivo)	Implemented & Integrated
Chemical Safety Guardrail	Implemented
PAE Expert Rerouting Queue	Implemented
WhatsApp History Dashboard	Implemented
Price Trend Prediction Model	Planned / Documented in Roadmap

20. TECHNICAL OBSERVATIONS & INSIGHTS

- High architectural modularity achieved via FastMCP and routing-controllers.
- Effective integration of Indic speech services (Sarvam AI) for regional voice accessibility.
- Robust domain safety enforcement preventing hazardous pesticide advice.

21. DOCUMENTATION LIMITATIONS

Audit results reflect static codebase inspection and local static testing. Live production cloud endpoints and private API keys were not executed against production external gateways.